

CLiC-it2026 · Palermo, Italy · September 14–16, 2026

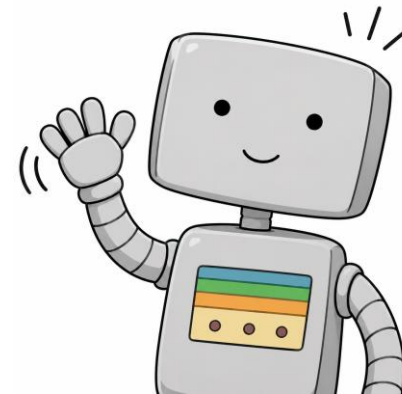
Lost in Transcription

Investigating the Role of Audio Information for Video-based Visual Statement Verification

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Motivation

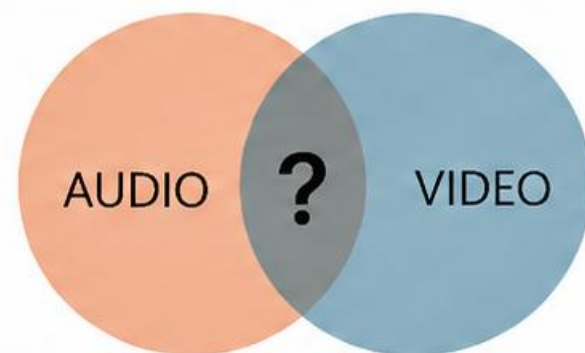
- Omnimodal LLMs have evolved to **process vision, language, and raw audio collectively.**
- Acoustic cues **enhance the visual experience.**
- Recent benchmarks have not isolate audio signal and **examine the specific causal role** that audio plays in forming the answer. [\(Fu et al., 2025; Geng et al., 2025\)](#)
- **Integration is not certain.**

?

How do the **two modalities interact** during reasoning?



Does audio **truly aid multimodal video** reasoning?





Research Question

RQ-1:

To what extent does raw audio affect video-based VSV when it is jointly processed with visual and textual information?

RQ-2:

How effectively do multimodal models integrate visual, textual, and auditory information when performing video-based reasoning and understanding tasks?

CONTRIBUTIONS

- 1 Extend MAIA with an auxiliary audio layer: transcribed content plus a coarse acoustic-context label for every video
- 2 Examine audio from two perspectives: as a raw signal using an omnimodal model, and as text through automatic transcription.
- 3 Run controlled experiments and targeted ablations to isolate how visual, textual and auditory signals interact

MAIA & the VSV Task:

- **MAIA**: native-Italian, video-based, competence-oriented benchmark
([Testa et al., 2025](#))
- **MAIA** was built around **visual grounding**, audio was never part of the original benchmark design
- 100 videos · 12 reasoning categories · 2,400 questions, each with a pool of 8 human answers
- **Visual Statement Verification** (VSV): given a video, pick the TRUE statement from a minimally-contrastive **True/False** pair

VSV PROMPT

<video> Choose the correct description:

A. {{ True Statement }}

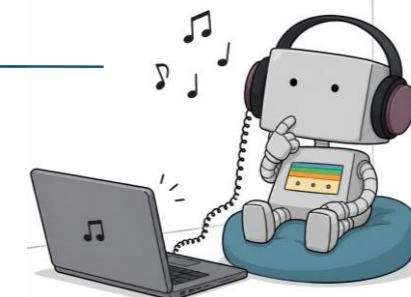
B. {{ False Statement }}

Answer only A or B.

EXAMPLE:

A. Nel video, il pizzaiolo appare concentrato mentre tira fuori la pizza dal forno.

B. Nel video, il pizzaiolo appare distratto mentre tira fuori la pizza dal forno.



Models & Setup:

Model	Role
Qwen2.5-Omni-7B	Main experiment with raw Audio (omnimodal)
Qwen2.5-VL-7B	Main experiment with Transcriptions (vision-language , no native audio)

Common settings:

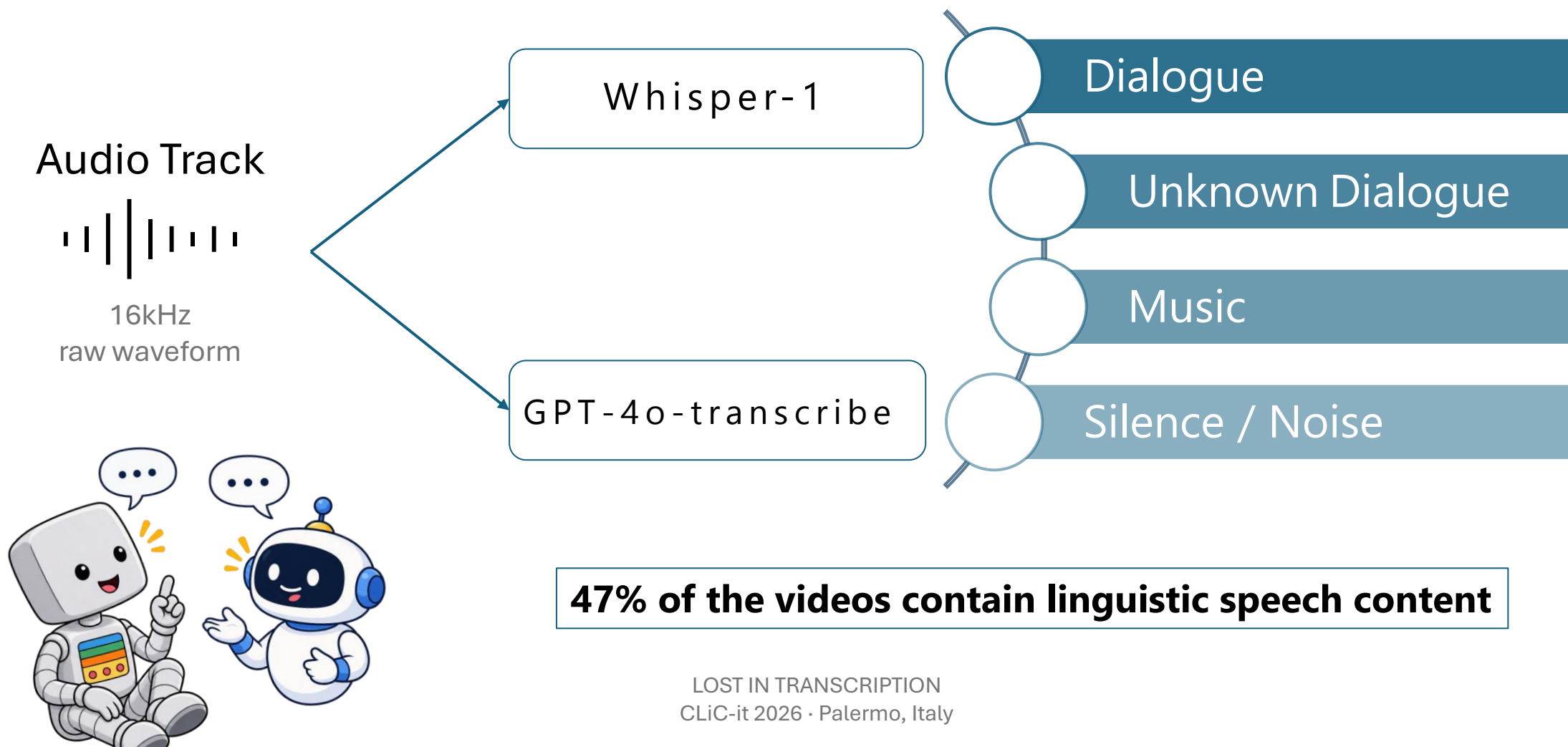


32 frames per second
uniformly sampled from every
video, across all experiments



Zero-shot inference, strict A/B
multiple-choice VSV prompt for
every configuration

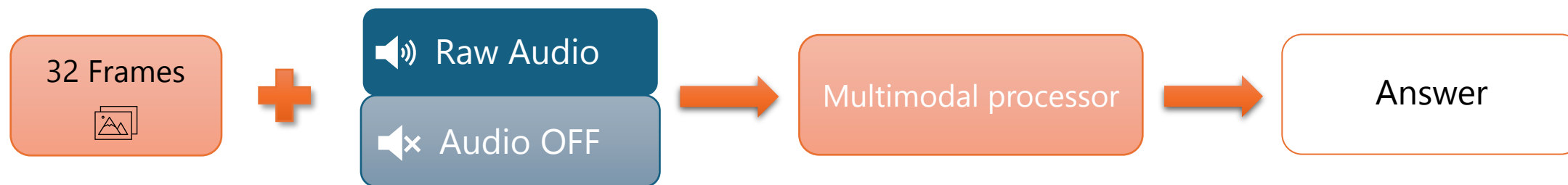
Building an Audio-Transcription Layer:



Experiment 1:

Raw Audio (Qwen2.5-Omni-7B)

*Every video is processed twice, once with audio, once without, so its effect can be **isolated video by video**.*



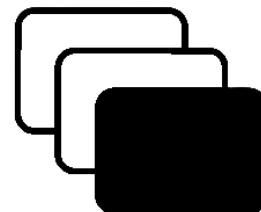
Ablation: Removing Visual Information

*Two visual-density settings, each tested with audio ON and OFF, **isolate how much audio matters as visual evidence shrinks***



Black Frame

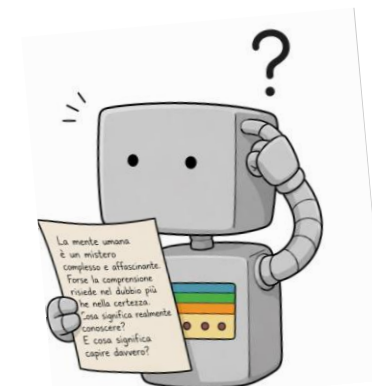
- Isolates pure audio
- No visual grounding is available at all
- Proxy for a "no-video" condition



1-Frame

- A minimal but non-zero visual anchor
- Removes temporal visual motion
- Tests the "static appearance" condition

Experiment 2: Transcriptions (Qwen2.5-VL-7B)



Unimodal Baseline

Linguistic-plausibility bias in the answer options alone

Transcription-only

Whether transcript text supports the decision on its own

32 Frames (w/o Transcription)

Visual grounding performance alone

32 Frames (with Transcription)

Whether the transcript adds anything once video is present

Visual Dominance

RAW AUDIO • QWEN2.5-OMNI-7B

-0.02 pp

82.70% → 82.68%

not significant • $p = .620$

No acoustic category is significant after Holm correction. Raw audio is effectively ignored when visual evidence is dense.

TRANSCRIPTION • QWEN2.5-VL-7B

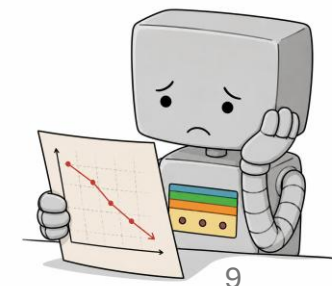
-1.89 pp

80.67% → 78.78%

highly significant • $p = 3.6 \times 10^{-17}$

Significant in every single acoustic category after Holm correction — a systematic effect, not a few outlier items.

Under dense visual evidence, **raw audio is neutral**, but **transcribed audio is not**.



RESULTS

The “Dialogue Paradox”

TRANSCRIPTS CAN BE WORSE THAN NO INFORMATION AT ALL

Unimodal Baseline (no video, no audio) **68.72%**

Transcription-only (transcript, no video) **63.98%**

Reading the transcript alone performs worse than guessing from the question's linguistic plausibility.

-4.74 pp
vs. baseline

EVEN CLEAR, INTELLIGIBLE SPEECH DOESN'T RELIABLY HELP

Dialogue category — raw audio -0.29 pp (n.s.)

Dialogue category — transcription **-1.16 pp (significant, p_Holm = .005)**

Even **understandable speech** can actively mislead **once visual evidence is already dense.**

Qualitative Case Studies

VIDEO 37 • LARGEST IMPROVEMENT

audio **+2.60 pp** (18 Wrong→Correct)

transcript **+1.56 pp** (8 Wrong→Correct)

A child falls and starts crying. Audio links the crying to the fall itself, not to a later, unrelated action — fixing a causal / temporal question the visual stream alone left ambiguous.

VIDEO 71 • LARGEST DROP

transcript **-5.73 pp** (15 Correct→Wrong)

audio **-4.17 pp** (12 Correct→Wrong)

A street vendor sells oranges. Audio is topically coherent (“oranges”, “beautiful and sweet”) but not fine-grained enough — it misleads the model on tool-use and timing questions.

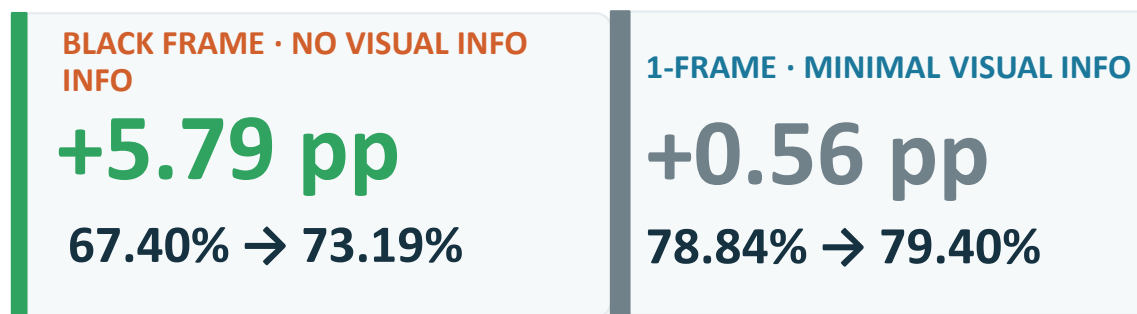
Audio helps only when it is directly **aligned with the exact discriminative detail** the task needs.

Attention Allocation & the Limits of Fusion



The model implicitly modulates audio's relevance to how much visual information is available.

But **more attention** doesn't mean **better reasoning**



One static frame is already enough to make audio's contribution **mostly vanish**.

Key Findings



**Visual Evidence
remains the
Dominant**



**Under dense Visual
Evidence:
raw audio $\approx$ no
effect**



**Transcriptions cause
a moderate but
systematic
interference**



Open question:

Is this a property of MAIA's visually-grounded design, or a general limitation of how current architectures gate modalities?

Powering down...

Thank you for your
human attention

DATA AVAILABLE ON GITHUB

